Master Board Power Tree and Buck Converter Analysis

Battery Management System for Electric Scooter | ECE 445 Team 35 | Technical Design Note

Design Objective

The master board must generate regulated rails for digital control, analog measurement, communication, and support circuitry from the scooter battery pack. The input range is approximately 18 V to 25.2 V, and the generated rails are 12 V, 5 V, and 3.3 V.

- The pack-to-12 V stage uses a synchronous buck converter for efficiency.
- The 12 V rail feeds LDO regulators for the 5 V and 3.3 V rails.
- LDOs are used after the buck stage because low-noise rails are valuable for ADC and sensing circuits.

Why a Buck Converter is Used First

A linear regulator from the full pack voltage to 12 V would dissipate the voltage difference as heat.

$$P_{loss,LDO} = (V_{in} - V_{out}) * I_{load}$$

$$P_{loss,LDO} = (25.2\ V - 12\ V) * 1\ A = 13.2\ W$$

13.2 W is excessive for a compact board regulator and would create a major thermal problem. The ideal LDO efficiency at this condition is also poor.

$$\eta_{LDO} = V_{out} / V_{in} = 12\ V / 25.2\ V = 47.6\%$$

A switching buck converter operating around 85%-95% efficiency greatly reduces heat generation during the large voltage step-down.

Why LDOs are Used After 12 V

After the efficient buck pre-regulator, LDOs generate the local 5 V and 3.3 V rails. This is acceptable because the downstream currents are lower, and the LDOs provide ripple rejection and lower output noise than additional switching stages.

$$P_{loss,5V} = (12\ V - 5\ V) * I_{5V} = 7 * I_{5V}$$

$$P_{loss,3.3V} = (12\ V - 3.3\ V) * I_{3.3V} = 8.7 * I_{3.3V}$$

The tradeoff is intentional: the buck handles the high-power conversion, while LDOs provide cleaner rails for MCU, ADC, and communication components.

Buck Inductor Ripple Analysis

The first design iteration used an inductor value that created excessive ripple current. For a buck converter, inductor ripple current can be estimated as:

$$\Delta I_L = V_{out} * (1 - D) / (L * f_s)$$

$$D = V_{out} / V_{in} = 12\ V / 25.2\ V = 0.476$$

$$\Delta I_{L,4.7uH} = 12\ V * (1 - 0.476) / (4.7\ uH * 500\ kHz) = 2.67\ A$$

With a 2 A nominal load, this ripple is too large because half the ripple exceeds the average current. That condition can force discontinuous conduction mode.

$$I_{L,avg} < \Delta I_L / 2 \Rightarrow \text{DCM risk}$$

Output Ripple and Corrected Inductor

Output voltage ripple increases with inductor ripple current and decreases with switching frequency and output capacitance.

$$\Delta V_{out} \approx \Delta I_L / (8 * f_s * C_{out})$$

Increasing the inductor to 10 uH reduced the ripple current to approximately 1.25 A, improving continuous-conduction operation, output stability, and downstream rail quality.

Delta I_L,10uH approximately 1.25 A

Design Case	Inductor	Ripple Current	Consequence
Initial	4.7 uH	2.67 A	High ripple, DCM risk, poor regulation.
Revised	10 uH	Approx. 1.25 A	Lower ripple, improved CCM behavior and output stability.

Measured Rail Verification

Rail	Requirement	Measured Value	Error
12 V	+/-5% of nominal	12.08 V	+0.67%
5 V	+/-5% of nominal	5.02 V	+0.40%
3.3 V	+/-5% of nominal	3.31 V	+0.30%